Week 04: two kinds of work

Programming task

Functions and practice.
Then the individual Midterm.
Practice and assessment stay separate.

Research task

After the Midterm: Research 3.
No research due during Week 4.
Discuss in Week 5 · proposed 5%.

Use this guide with the same Week notebook; do not create a second set of weekly tasks. Email: submit.progmis@gmail.com.

Programming practice

Functions, arguments and returned values
Predict → run → change → repair → explain

Three core tasks: understand the Python

Predict a return

Trace `rectangle_area(3, 4)`. Separate definition from call.

Change and reuse

Call with (3, 2). Add 1 to the returned value. Predict both.

Repair the function

The function prints but returns None. Fix the returned value.

Open Week_04.ipynb → Session 2. Keep a prediction, result and explanation for each task.

Core 1–2: predict, then change one thing

```
01 def rectangle_area(width, height):
02     return width * height
03
04 print("defined")
05 result = rectangle_area(3, 4)
06 print(result)
```

1. Separate defining the function from calling it.
2. Call (3, 4): 12; call (3, 2): 6.
3. Use the returned 6 in `result + 1`: 7.

Use the complete Core 1 and Core 2 starters in the notebook; explain each changed result.

Core 3: find and repair one mistake

```
01  def rectangle_area(width, height):
02      print(width * height)  # Deliberate missing-return bug.
03
04  result = rectangle_area(3, 4)
05  print("caller received:", result)
06  # Repair the function and check both returned values.
```

1. Predict the printed area and the caller’s result.
2. Repair print-only behavior so the function returns.
3. Retest returned values for (3, 4) and (3, 2).

Edit the matching Core 3 cell. Keep the wrong result, your repair and a retest.

Done: explain the concept, then save

Step	What to do / keep
Program	Required function and returned dictionary
Tests	All four common cases with all three fields asserted
README and ZIP	Purpose, commands, expectation, limitation and AI disclosure
Own explanation	Be ready to trace a case in your announced individual slot

After practice, explain arguments and return values. Stop at the Midterm start time.

How the Midterm fits

Begin in Session 2; continue in Session 3
Implement, test, package and review

Midterm: follow the existing notebook task

Step	What to do / keep
Open / implement	Midterm begins → `summarize_request(lines, reviewed)`.
Required rule	Unreviewed: hold; else >=20: team; else: standard. Return totals too.
Four tests	A: 4/20/hold; B: 3/9/standard; C: 4/20/team; D: 6/22/team.
Package	midterm.py + test_midterm.py + README.md; include AI disclosure.
Keep / explain	Fresh-folder test result, ZIP and backup; oral slot announced.

Use the complete input table in the notebook. Threshold 16 and main/import detail stay optional.

Research input validation

What to research

Find out why programs check input before using or saving it.

Possible directions

Choose one field and an invalid value. Consider the allowed values, the message to show and what should remain unchanged.

A short summary example

A container count must be a positive integer. If the input is “many”, ask for a whole count and leave saved stock unchanged.

Discuss in Week 5 · summarize in your own words and cite your sources.